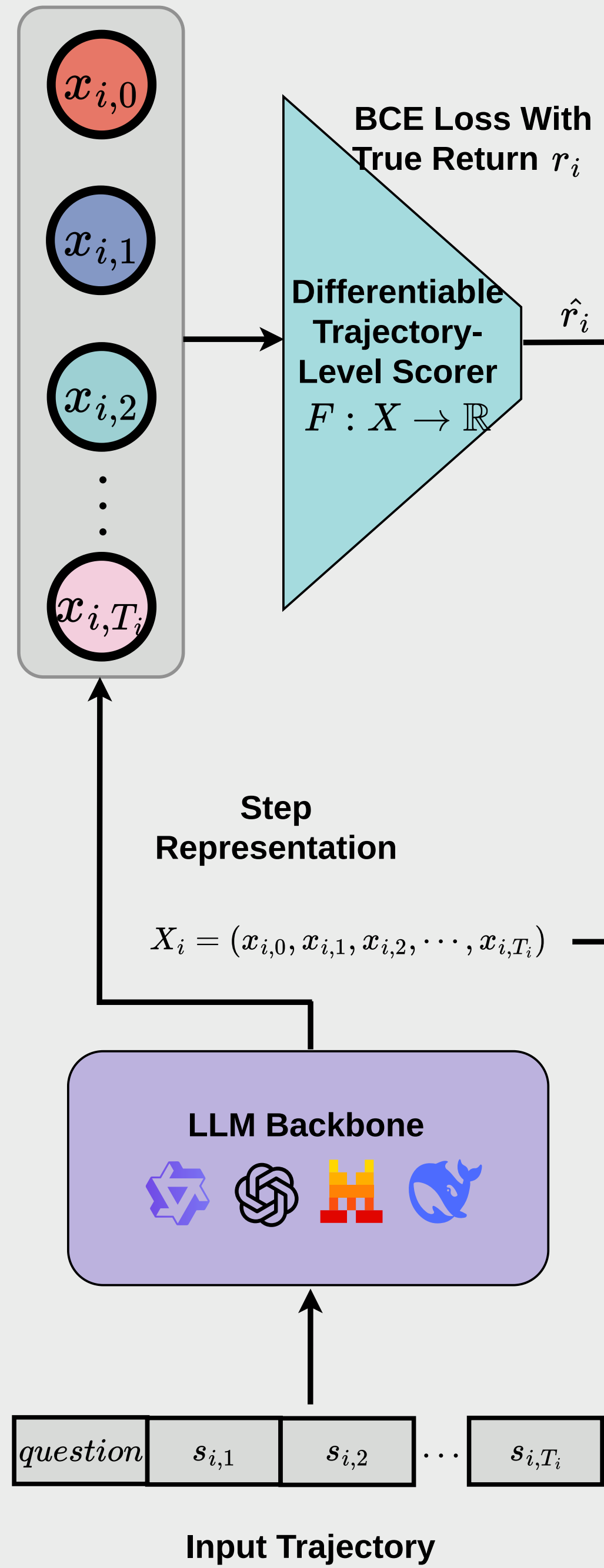


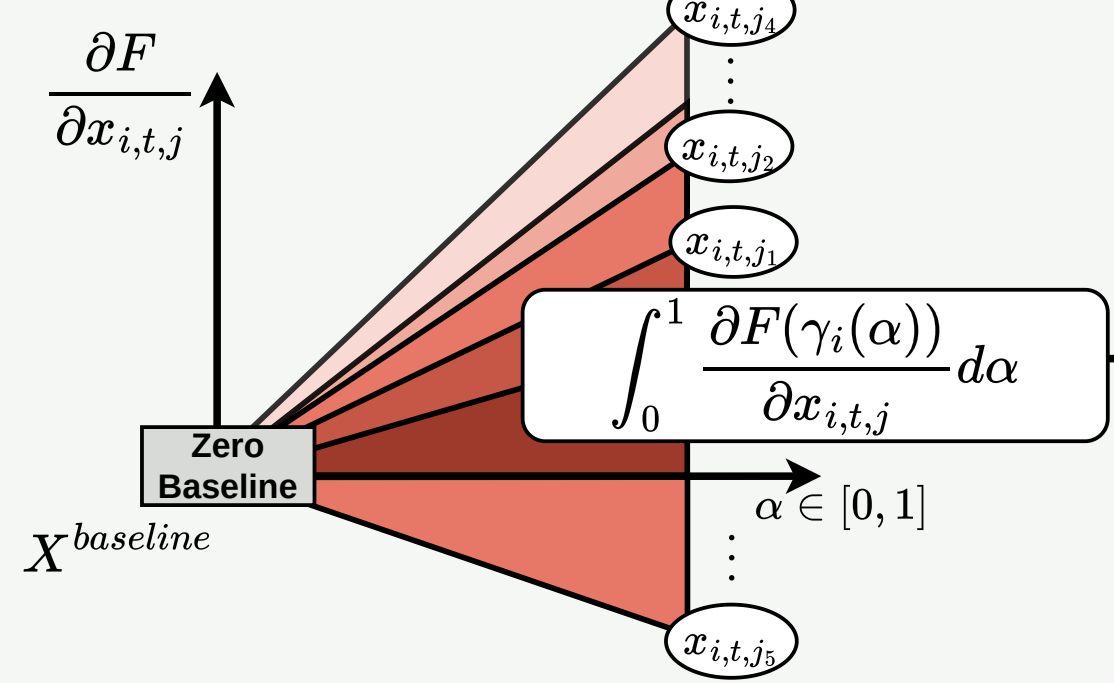
Learning the Trajectory Scorer



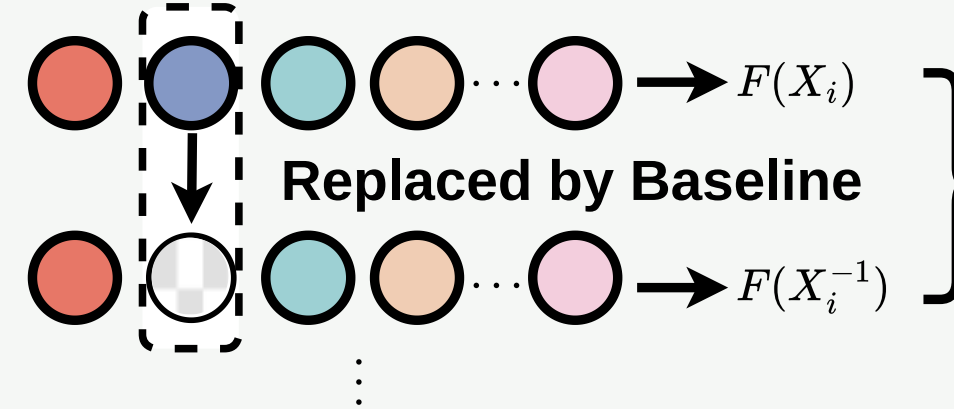
Construct Step-Level Labels

Local Signal Construction

Aumann-Shapley IG



Counterfactual Ablation



Causal Structure Signals

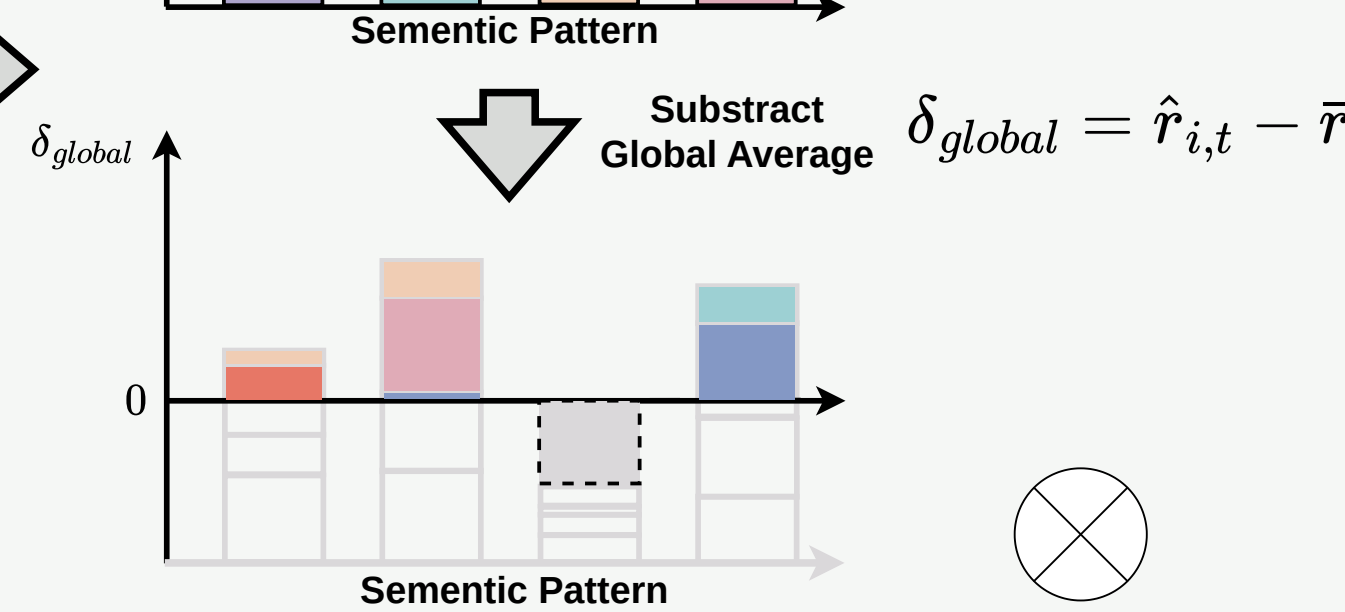
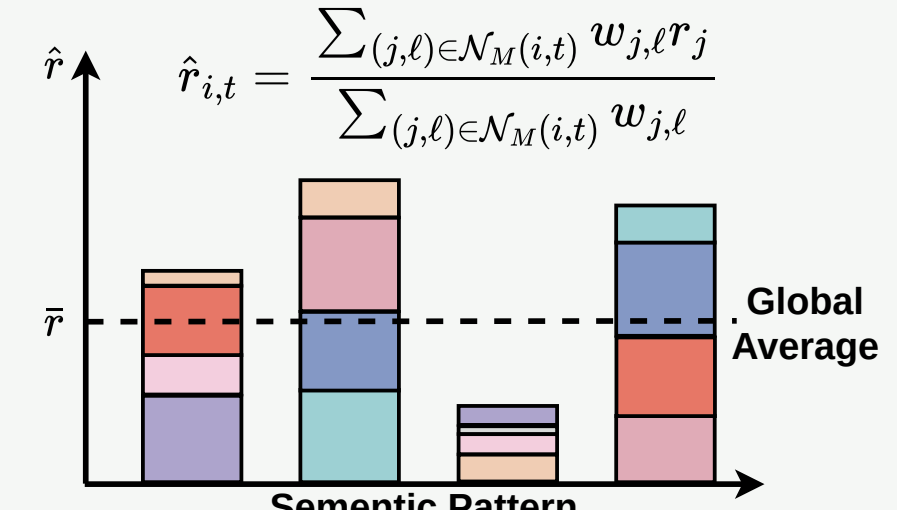
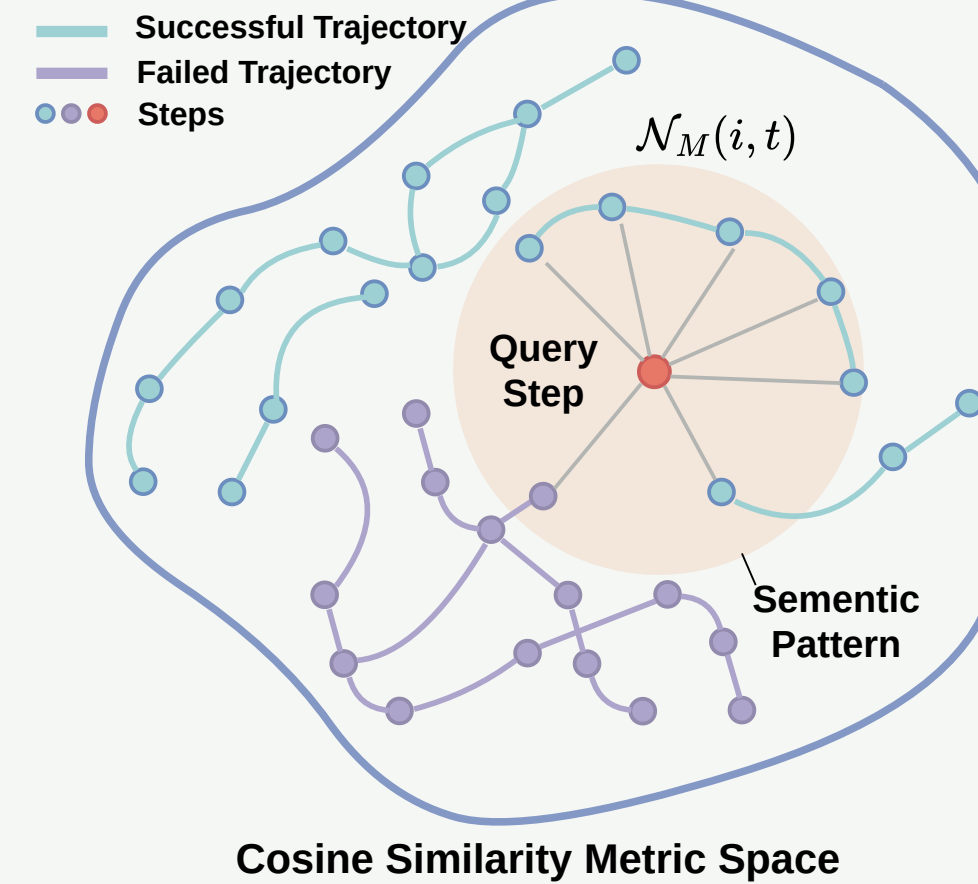
$$R_{i,t}^{causal} = F(X_i^{-t}) - F(X_i)$$

Step-Level Aggregation

Consistency Filter

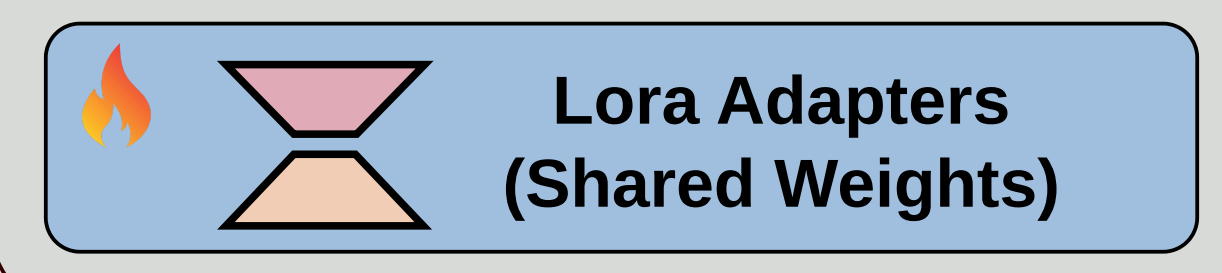
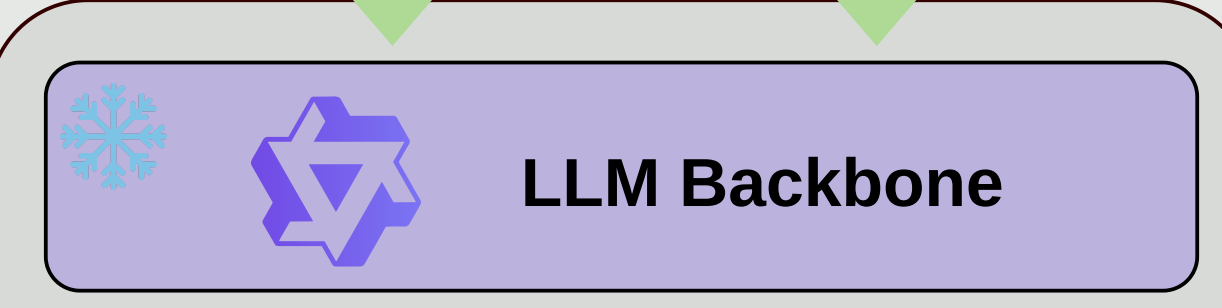
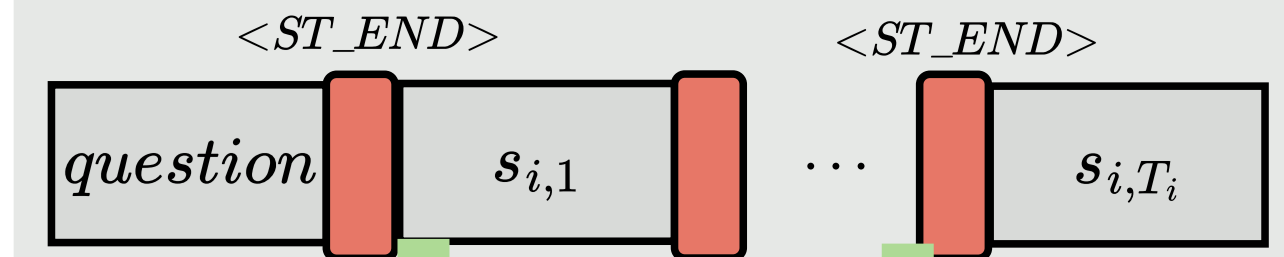
δ_{local}

Global Experience Prior

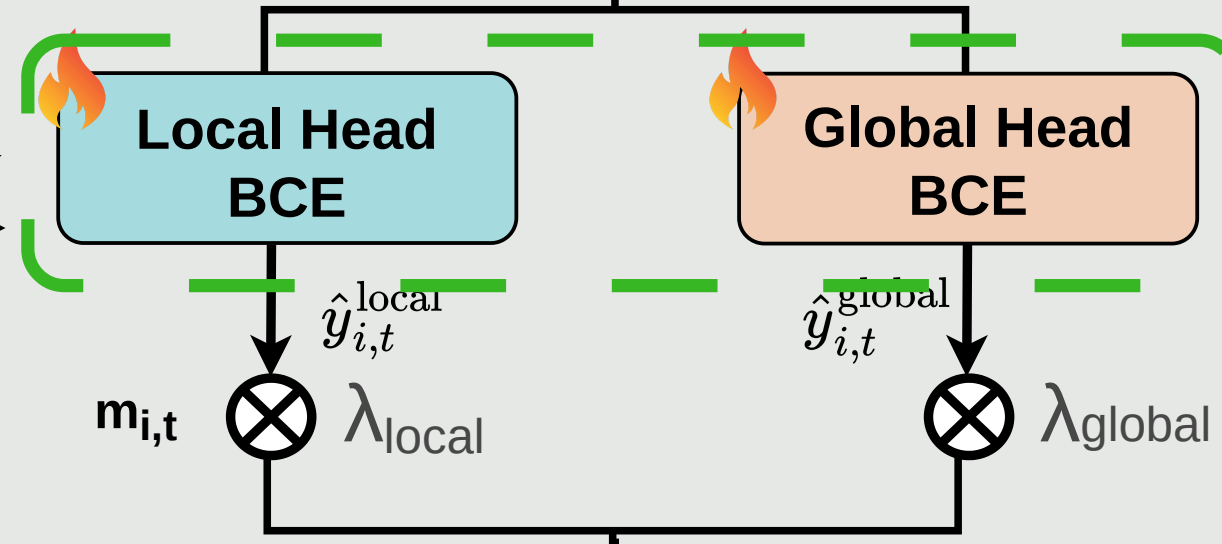
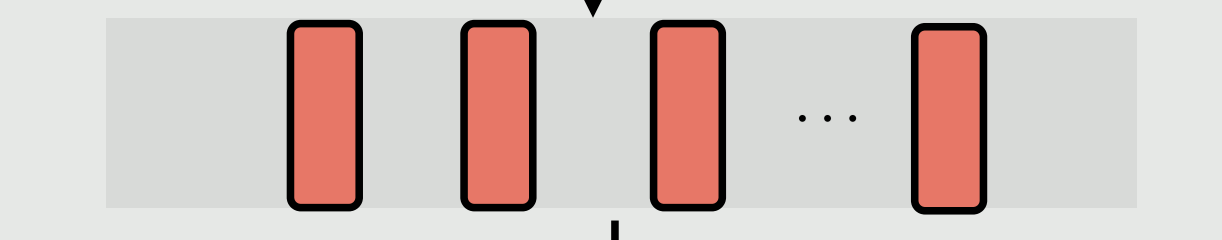


Dual-Head Training

Input Sequence with **<ST_END>**

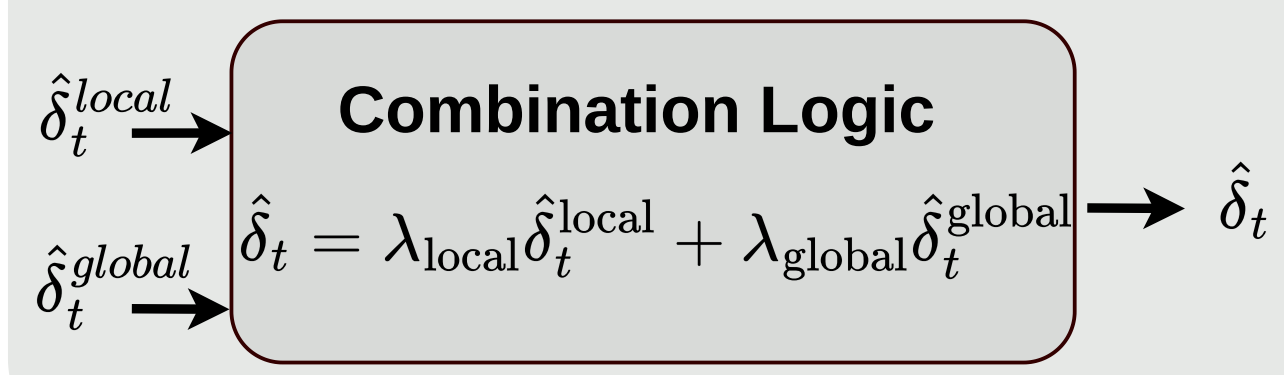


<ST_END> for Scoring



$$\mathcal{L}_{PRM} = \mathbb{E}[\lambda_{local} \sum_t m_{i,t} BCE_{local} + \lambda_{global} \sum_t BCE_{global}]$$

Inference Time (Process Reward Calculatin)



Final Process Reward